

# Kiran Naregal

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## PROFESSIONAL SUMMARY

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AI Engineer with production experience building LLM-powered document processing pipelines, AI agent systems, automated code review workflows using LangChain and Hugging Face, backed by expertise in Python, cloud-native serverless architectures, and full-stack development.

## PROFESSIONAL EXPERIENCE

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### AI Engineer

July 2025 – Present

*Vayavya Labs, Bangalore*

- Architected an end-to-end Python pipeline for converting hardware specification PDFs to structured Markdown, utilizing Docling library with OCR models for accurate text extraction from complex technical documents.
- Integrated LLM-powered automatic captioning system to generate descriptive text for figures, diagrams, and tables, enhancing semantic understanding for RAG-based retrieval systems.
- Implemented document parsing, content structuring and enabling efficient indexing of technical documentation for AI-powered knowledge bases.
- Optimized pipeline performance through iterative refinement and cross-functional collaboration, delivering high-quality structured outputs with 97-98% accuracy for downstream LLM processing and RAG-based retrieval applications.
- Developed an automated C assessment review system using LangChain that processes C source files and generates structured review reports in Excel format, significantly reducing manual code review effort and accelerating the training evaluation process.
- Contributing to Tvastaa VP, an agentic workflow built with the Claude Agent SDK that generates SystemC models from IP datasheets by chaining into the existing Docling-based preprocessing pipeline; built the Python harness, tool integrations, and orchestration layer, while collaborating with the SystemC domain team on prompt design and validation.
- Helped cut internal SystemC model development time from 4-6 months to 2-3 months; the workflow was demoed publicly at DVCon 2026, with client deployment planned in the coming months.

### Student Trainee

Feb 2024 – May 2024

*BGSW, Bangalore*

- Architected serverless workflows using AWS step functions workflow studio and lambda function for a subscription-based platform. Built and deployed backend services utilizing lambda for serverless computing.
- Developed Python-based Lambda functions for backend services, implementing automated email notification middleware through orchestrated workflows.

### Machine Learning Intern

Jan 2023 – May 2023

*Dana Anand, Hubli*

- Trained a robust object detection model to accurately identify whether each manufacturing step was completed. The model was deployed for real-time inference, providing immediate feedback and alerts.
- This proactive system effectively minimizes errors and prevents unfinished products from reaching customers, enhancing overall product quality and customer satisfaction.

## EDUCATION

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### B.E in Computer Science

Jan 2021 – June 2024

*KLE Technological University, Hubli*

CGPA: 8.76

## PROJECTS

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### Code Review Agent

[GitHub](#)

- Built a multi-agent code review API with LangGraph orchestrating three specialist agents (logic, security, style) and a supervisor synthesis node, backed by a provider-agnostic LangChain LLM layer supporting Anthropic, OpenAI, and Ollama via a single environment variable, with a vanilla JS frontend supporting three input modes (code snippet, file upload, GitHub URL) and severity-tagged findings with copy/download report.

- Designed the full system architecture from scratch — FastAPI backend with structured Pydantic validation at the LangGraph TypedDict boundary, rotating file logging, and a dark-themed single-page frontend with no framework dependencies, served as three static files.

### DGX Server Management Portal

- Developed an internal web-based portal for DGX server allocation and scheduling, supporting up to 150 student users with real-time server availability tracking and future booking functionality.
- Built an admin dashboard for centralized GPU resource management, enabling server user assignment, maintenance status updates, and streamlined coordination of server access requests.

### TECHNICAL SKILLS

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- Languages: Python, JavaScript, C/C++
- Python Libraries & AI Frameworks: Docling, Pandas, PyPDF2, LangChain, LangGraph, Hugging Face Transformers, FastAPI, Claude Agent SDK, Claude Code
- Cloud Platforms: AWS (Lambda, Step Functions, S3, EC2, CloudWatch)
- Web Technologies: React, Node.js, Express, Next.js
- Databases: MongoDB, SQL
- Tools & DevOps: Docker, Git, GitHub
- Frameworks: Tailwind CSS

### ACHIEVEMENTS

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- Published a paper on groove detection at the ICDSIA conference.

### CERTIFICATIONS & COURSES

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- MongoDB Node.js Developer Path [Certificate](#)
- The Ultimate React Course 2024: React, Next.js, Redux & More [Certificate](#)
- Building with Claude API [Certificate](#)